Sangam Panchal

Java Backend Engineer | Spring Boot, REST APIs, PostgreSQL
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Java Backend Engineer with hands-on experience building RESTful APIs and payment-processing systems using Java 21, Spring Boot, Spring Data JPA, Hibernate, and PostgreSQL. Skilled in concurrency-safe transaction design with optimistic locking, Spring Security authentication and authorization, and third-party API integration (Stripe). Experienced deploying production applications on Oracle Cloud Infrastructure (OCI) with Nginx reverse proxy, Neon PostgreSQL, and Vercel, following 12-Factor App practices. Also built an AI agent in Python using Gemini Function Calling with deterministic security guardrails.

TECHNICAL SKILLS

Languages: Java 21, Python, SQL
Backend Frameworks: Spring Boot, Spring Framework, Spring Data JPA, Spring Security, Hibernate (ORM), Maven
API Development: RESTful APIs, REST API Design, Stripe API Integration, Postman
Databases: PostgreSQL, Neon Serverless PostgreSQL, SQLite, MongoDB
Cloud and Deployment: Oracle Cloud Infrastructure (OCI Compute), AWS (EC2, S3), Vercel, Nginx (Reverse Proxy), Linux (Fedora)
Architecture and Concepts: Optimistic Locking, Event-Driven Architecture, Transaction Management, Role-Based Access Control, 12-Factor App Methodology, Payment Processing
AI and Agentic Systems: Gemini Function Calling, Tool Orchestration, Deterministic Guardrails (AST, Regex)
Tools: Git, GitHub, Maven, Postman

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering | 2021 - 2026
R.D. Engineering College, Ghaziabad, Uttar Pradesh, India

PROJECTS

MeshPay Backend System | Java 21, Spring Boot, Spring Data JPA, Hibernate, PostgreSQL
GitHub Repository
- Designed and built a payment routing and transaction backend in Java and Spring Boot, exposing 10 RESTful API endpoints with Spring Data JPA and automated Hibernate schema management on PostgreSQL.
- Implemented concurrency-safe transaction processing across payment routes using optimistic locking (@Version) to prevent conflicting updates.
- Automated database seeding on startup with an ApplicationReadyEvent listener (event-driven lifecycle).
- Followed the 12-Factor App methodology by moving database configuration and secrets into environment variables.

NexCart E-Commerce Platform | Java, Spring Boot 3, Spring Security, Stripe API, Neon PostgreSQL, OCI, Nginx, Vercel
GitHub Repository | Live Demo
- Developed a modular e-commerce backend in Spring Boot 3 integrated with the Stripe SDK, supporting dynamic multi-currency checkout and end-to-end payment lifecycle handling.
- Secured 12+ REST API endpoints with Spring Security, implementing role-based authorization and session security.
- Eliminated duplicate orders through server-side inventory validation.
- Deployed a decoupled production architecture: Spring Boot backend on Oracle Cloud Infrastructure (OCI Compute) behind an Nginx reverse proxy, with Neon PostgreSQL as the database and a Vercel-hosted frontend.

AutoGov-DB: Autonomous Database Governance Copilot | Python, Gemini Function Calling, SQLite, SQL, AST
GitHub Repository
- Built an autonomous AI agent in Python using Gemini Function Calling to inspect infrastructure cluster metrics and correlate incidents across multiple relational tables.
- Developed a deterministic security guardrail layer using AST parsing and regex interceptors that blocks DDL/DML mutations (DROP, DELETE, UPDATE) and semicolon injection attempts before query execution.
- Implemented runtime schema discovery with a get_schema() tool, enabling the agent to generate multi-table SQL JOIN diagnostics autonomously without rigid prompting.

CERTIFICATIONS

Certificate Course on Data Analytics | ICT Academy and Capgemini (CSR Initiative) | December 2024
Focus: Data Analysis, Data Handling, Analytical Thinking